

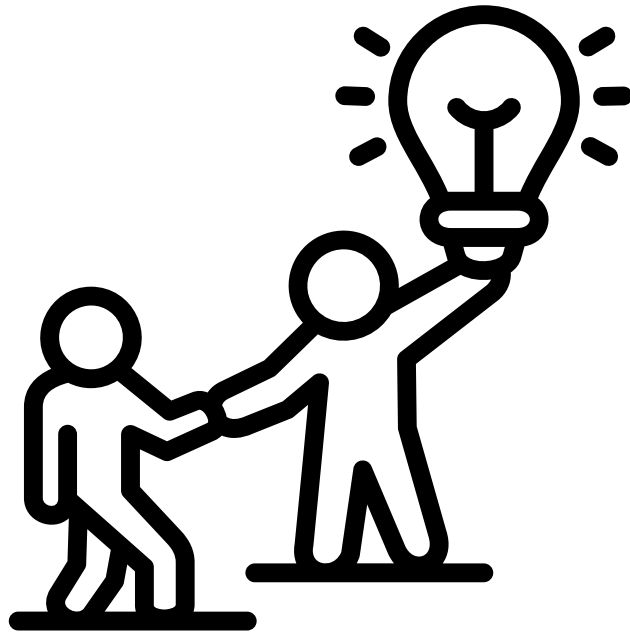
2026 EN813701 Web Application Development

AI Line Chatbot for Agricultural Drone Knowledge Base

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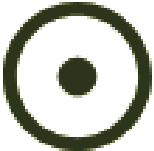


Motivation



- We want to help farmers or those who use drones for agriculture to easily find important information.
- We want to learn how to use RAG to create a Line Chatbot.

Related Works

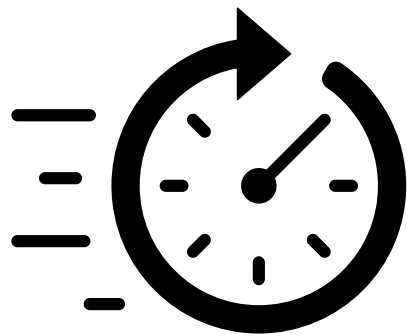
Work	Similarity	Difference
AgriChat  AgriChat	AI Chatbot for Agriculture	The scope is broad and not specific to agricultural drones.
LoginEKO  LoginEKO	AI Chatbot for Agriculture	The scope is broad and not specifically Thailand.
UAS Portal CAAT 	Provides information about agricultural drones.	Users must search for information themselves.

Objectives



1. Develop a LINE chatbot that can answer a wide range of questions.

- 80% of the test questions have answers that can be traced back to their original sources.



2. Designed for fast response.

- 70% of responses take an average of 5-10 seconds.



3. It is genuinely beneficial to farmers/drone service providers.

- Collect satisfaction survey results (scores 1–5).
From a user.

Scope Features

Feature	MVP
1. Guidelines for using drones for different types of plants.	The focus is primarily on rice because it has a larger user base than other types of crops.
2. Preparing to spray chemicals.	Provides accurate information for safe chemical preparation and spraying.
3. CAAT's laws and regulations regarding drones.	Let's start by providing information on the most important basic regulations of CAAT.
4. Troubleshooting and maintenance.	Provides technical information, maintenance instructions, and troubleshooting capabilities, all in Thai.

Tools & Technology

Frontend / Backend



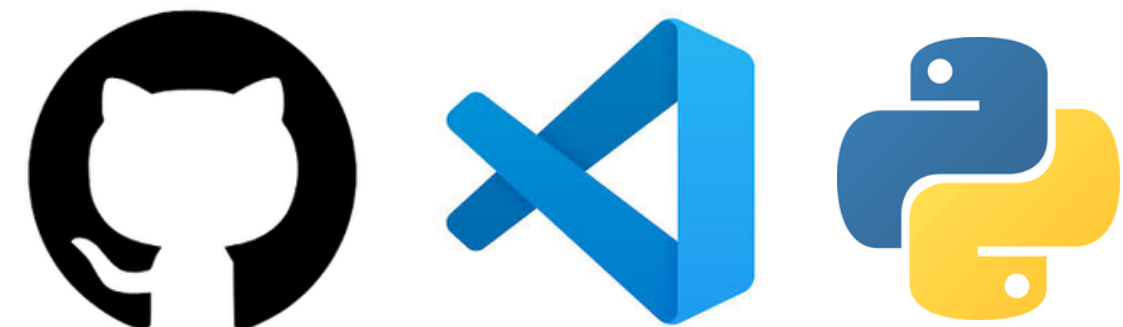
Database



LLM



Development



Persona

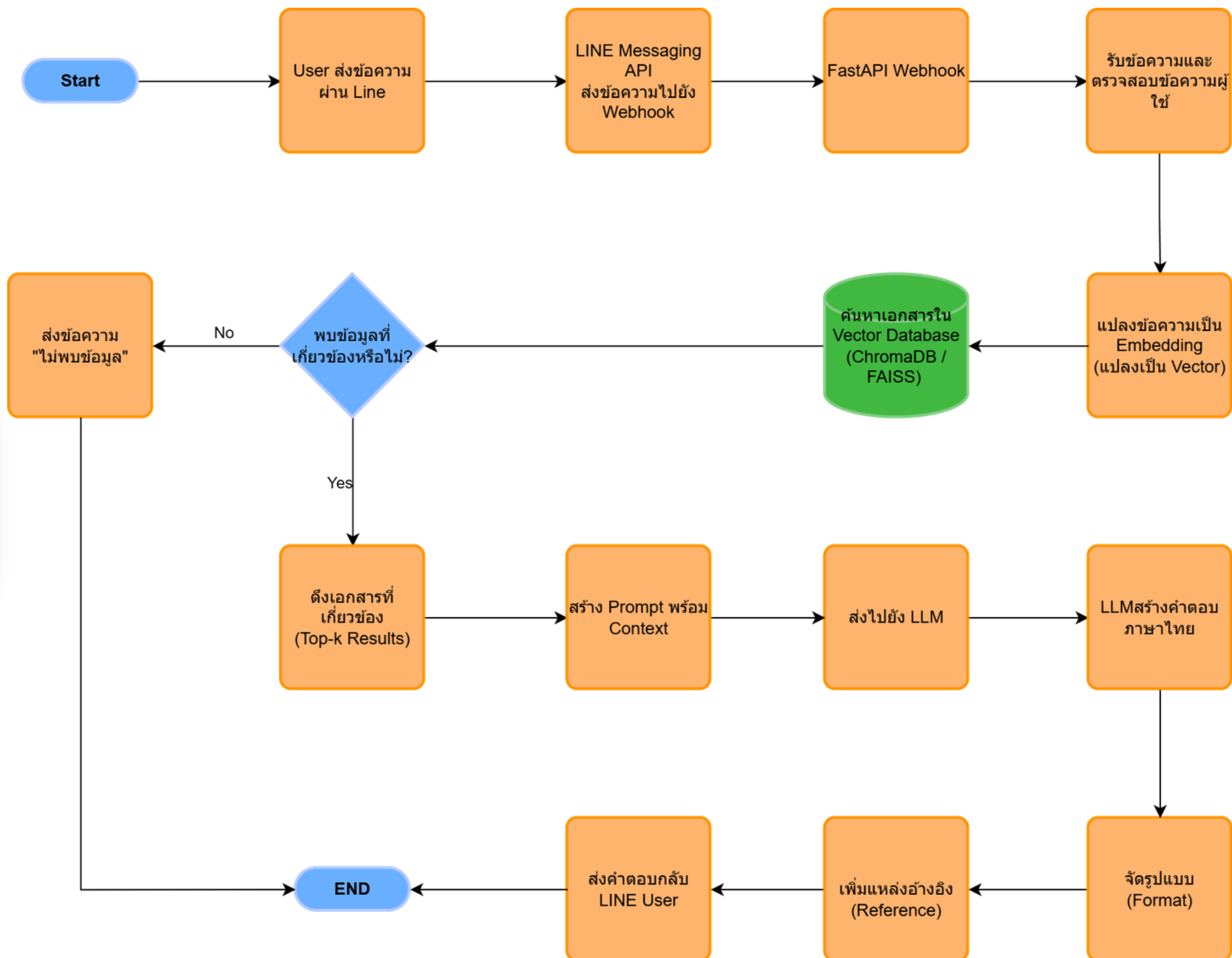


A farmer who wants to know about drone laws but is having trouble finding the information he needs by browsing the web.

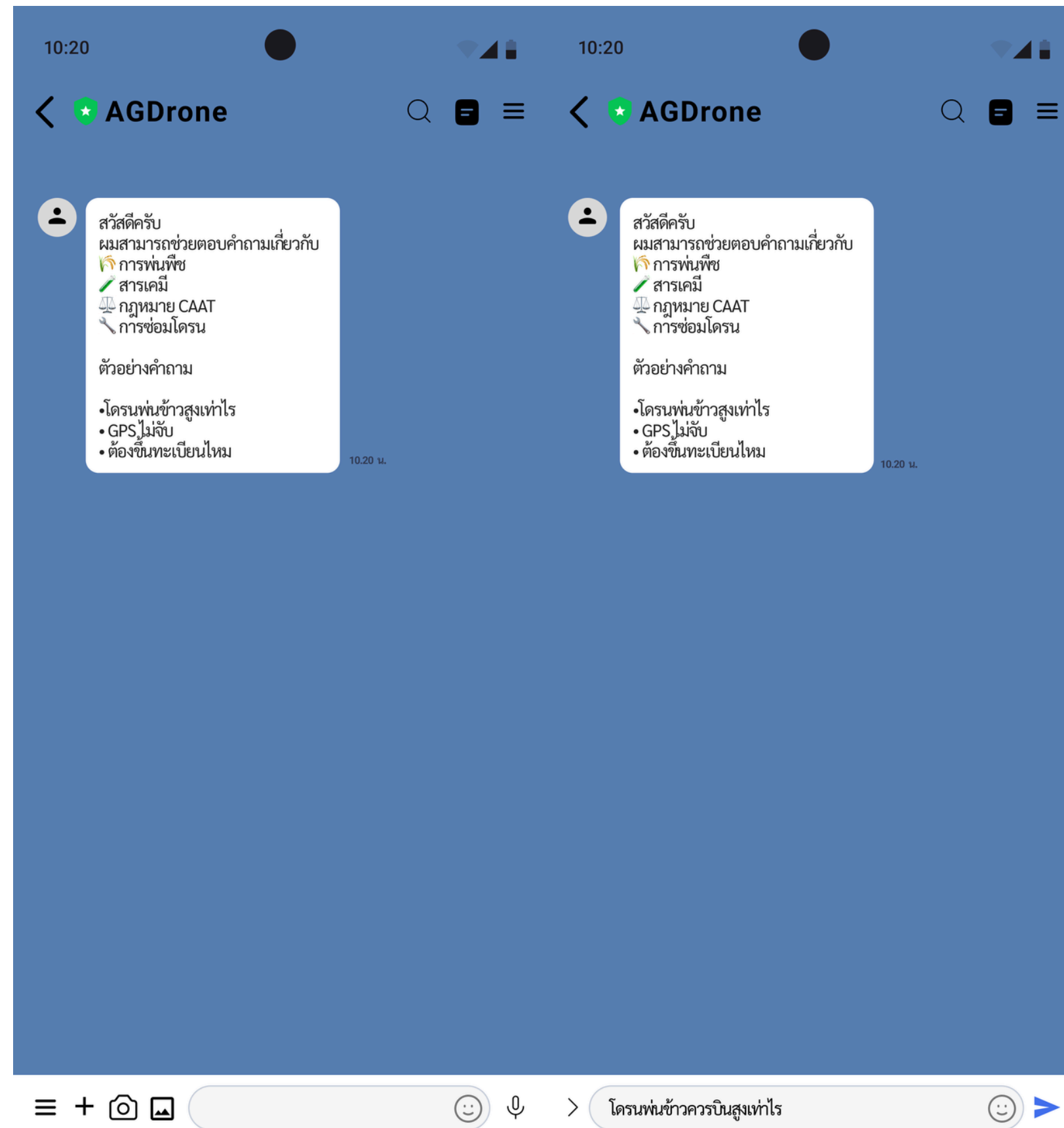


A drone service provider who wants to study the correct way to prepare chemicals in drones for agricultural use.

Flow Chart



User Interface



First Page

Type Question

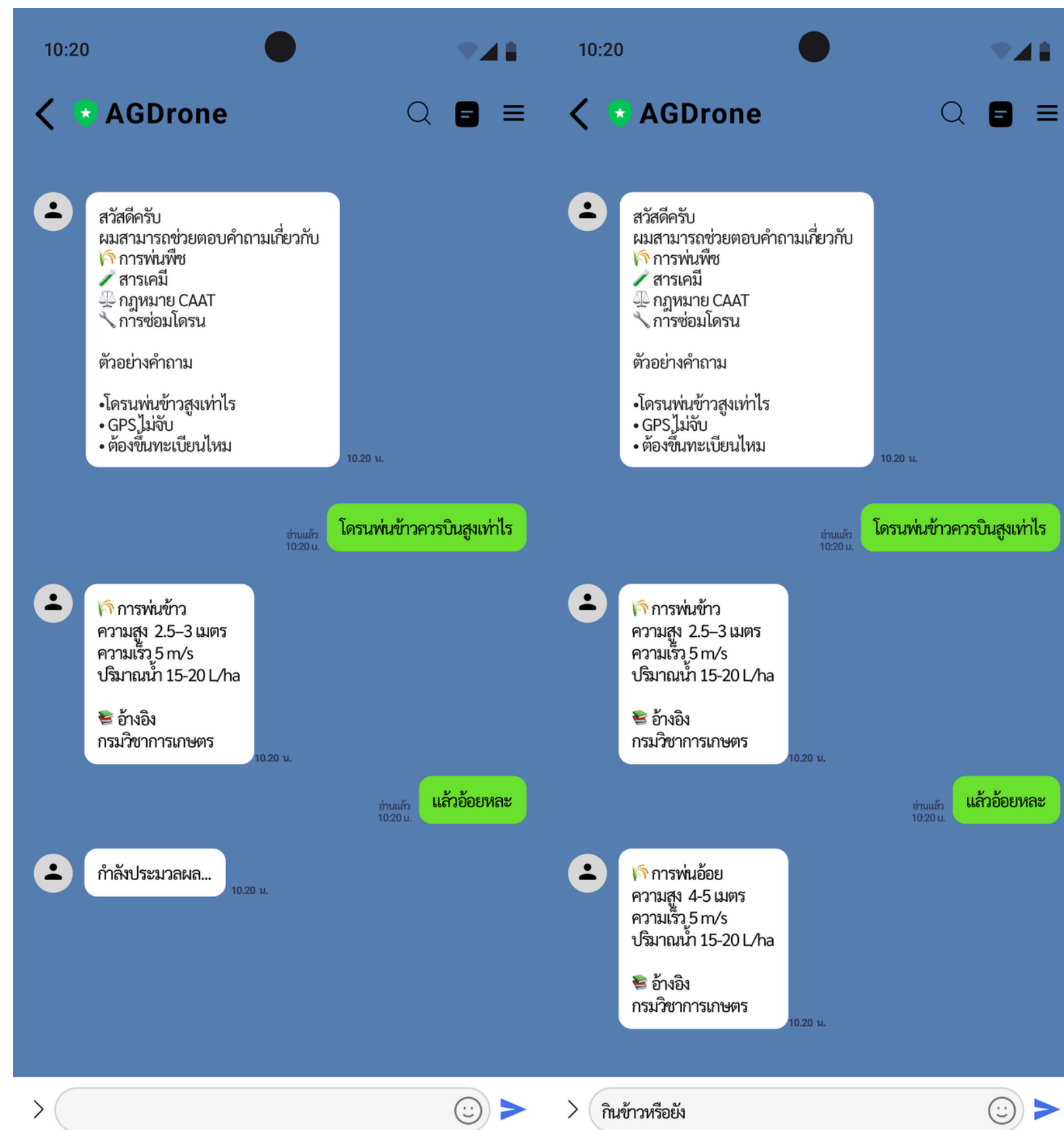
User Interface



Processing

Answer

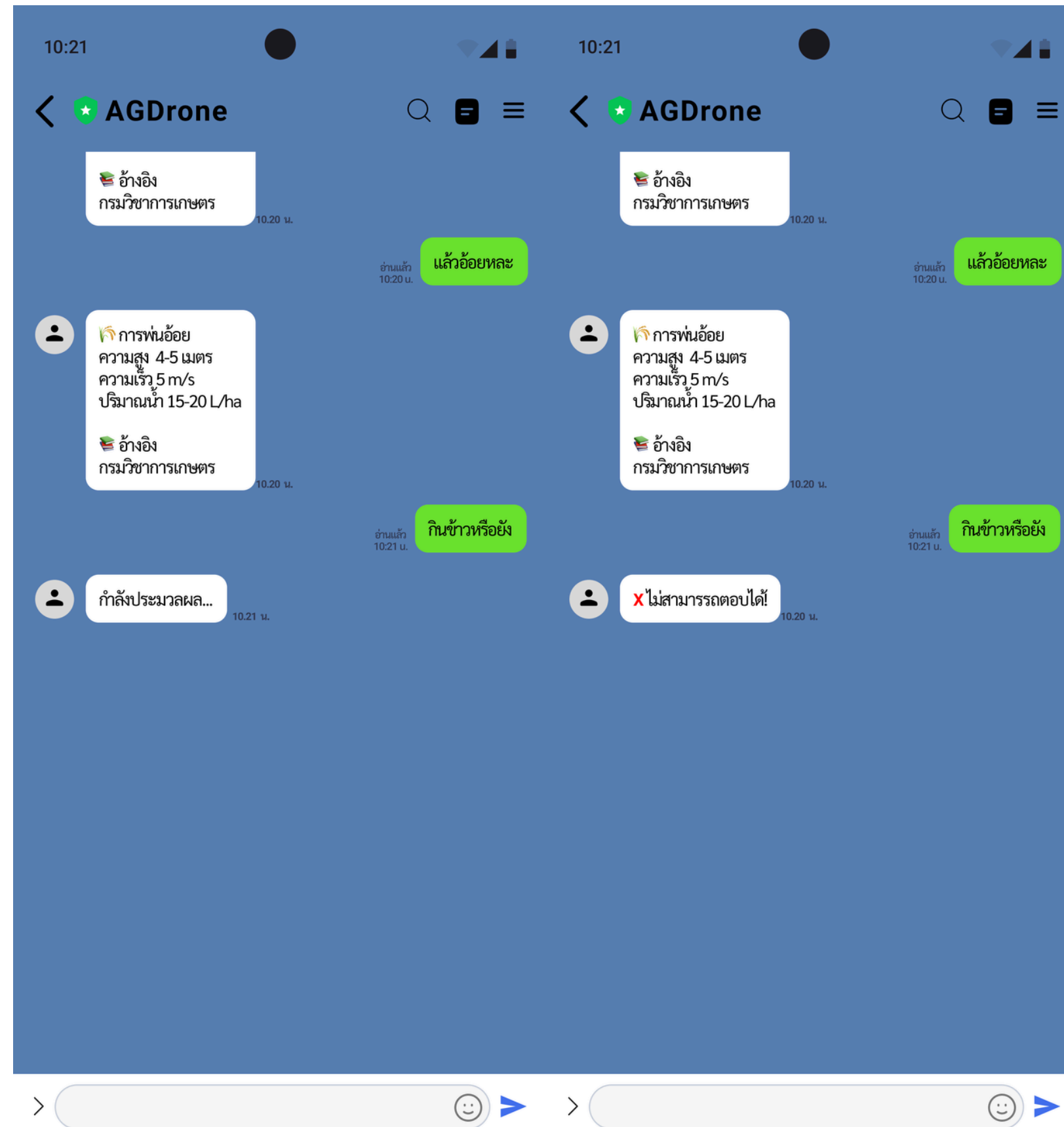
User Interface



Context

Answer Context

User Interface



Bad Question

Send Can't Reply

API Integration Strategy



- LINE Messaging API

Convert and send messages from farmers in JSON format.



- FastAPI Webhook

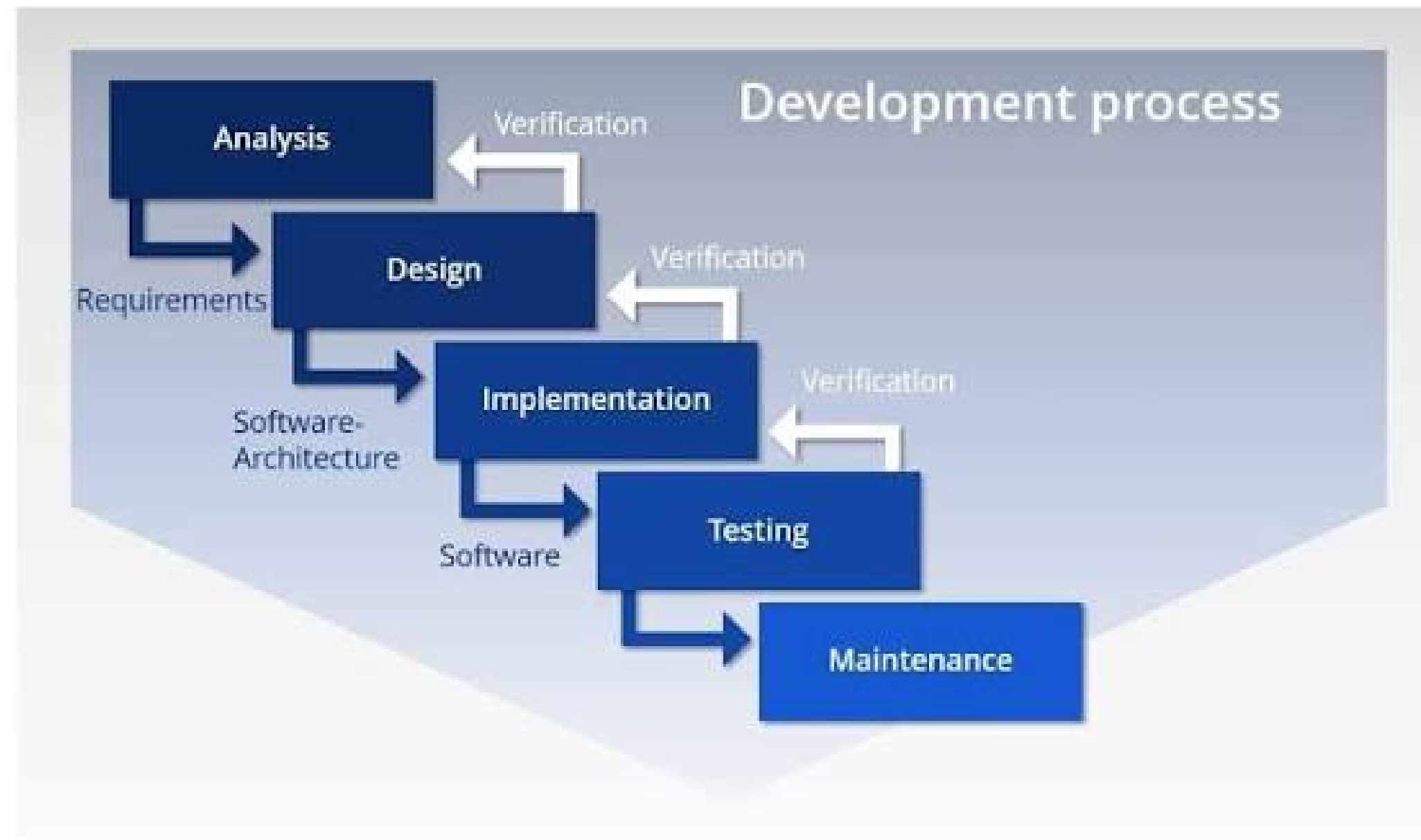
Send a message to Vector Database to retrieve the relevant information.



- LLM API

Gather facts, sources, and respond in Thai.

Methodology



Waterfall

Timeline

Activity / Week	W1	W2	W3	W4	W5	W6	W7	W8
Define and study the topic; analyze the problem.	✓							
Create the database and schema.		✓						
Develop Backend / Webhooks			✓	✓				
Develop the RAG system.					✓			
Develop a logic system to prevent errors.						✓		
Test,improve the system and report.							✓	✓

Conclusion

- Expected outcomes: An AI LINE Chatbot system to assist farmers or related parties who use drones.
- Success metrics: The chatbot can comprehensively answer specific questions about agricultural drones and quickly.

References

Agrichat <https://www.agrichat.ai>

LoginEKO <https://www.logineko.com>

UAS Portal CAAT <https://www.uasportal.caat.or.th>

Claude <https://www.claude.ai>